

LABORATORY OBSERVATION / RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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Section: B

1. TITLE

HTML5 Graphics and CSS3 Web Styling

2. AIM

To understand how CSS3 enhances the appearance of web pages and to explore HTML5 graphics using the Canvas element by designing a visually styled webpage.

3. OBJECTIVE

- To draw basic shapes and text on a webpage using the HTML5 Canvas element.
- To understand CSS3 syntax and different types of selectors.
- To apply colors, fonts, borders, margins and padding to page elements.
- To use CSS3 backgrounds, gradients, box-shadow, text-shadow and rounded corners.
- To implement hover effects on buttons and navigation links.

4. THEORY

The HTML5 <canvas> element provides a drawable region on a webpage that can be controlled using JavaScript. Its 2D context object exposes methods such as strokeRect(), arc(), moveTo()/lineTo() and fillText() to draw rectangles, circles, lines and text respectively.

CSS3 (Cascading Style Sheets, Level 3) is used to control the presentation of HTML elements. Selectors (element, class, id) target specific elements, while properties such as background, border-radius, box-shadow, text-shadow and linear-gradient() control their appearance. The :hover pseudo-class applies styles only while the mouse pointer is over an element, and combined with the transition property it produces smooth animated effects such as a card lifting or a button changing color.

5. ALGORITHM / PROCEDURE

1. Create a new HTML file and declare the HTML5 document type.
2. Add a <canvas> element inside the body and give it an id.
3. In a script block, get the 2D drawing context of the canvas using getContext("2d").
4. Draw a rectangle using strokeRect().
5. Draw a circle using arc() with beginPath() and stroke().
6. Draw a straight line using moveTo() and.lineTo().
7. Display the student's name using fillText().
8. Build the page layout with a header, navigation bar, three content cards and a footer.
9. Apply CSS3 background colors and a gradient background to the header.
10. Style the cards with borders, border-radius, padding, margin and box-shadow.
11. Add a text-shadow to the header title and apply different font families.
12. Add hover effects on the navigation links and on the buttons using :hover and transition.

13. Save the file, open it in a browser and verify the drawing and styling.

6. PROGRAM

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My Styled Webpage</title>
</head>
<body>
  <header>
    <h1>My Styled Webpage</h1>
    <p>
      Learning HTML5 Canvas and CSS3 Web Styling
    </p>
  </header>
  <nav>
    <a href="#home">Home</a>
    <a href="#canvas">Canvas</a>
    <a href="#cards">Services</a>
    <a href="#contact">Contact</a>
  </nav>
  <main>
    <section id="canvas">
      <h2>HTML5 Canvas Graphics</h2>
      <canvas id="myCanvas" width="700" height="350">
        Your browser does not support HTML5 Canvas.
      </canvas>
    </section>
    <section id="cards">
      <h2>Our Learning Areas</h2>
      <div class="card-container">
        <article class="card">
          <h3>HTML5</h3>
          <p>
            HTML5 is used to create the structure
            and content of a webpage.
          </p>
          <button>Learn HTML</button>
        </article>
        <article class="card">
          <h3>CSS3</h3>
          <p>
            CSS3 is used to add colours, fonts,
            spacing, borders, and visual effects.
          </p>
          <button>Learn CSS</button>
        </article>
        <article class="card">
          <h3>HTML5 Canvas</h3>
          <p>
            Canvas is used to draw shapes, text,
            graphics, animations, and games.
          </p>
          <button>Explore Canvas</button>
        </article>
      </div>
    </section>
  </main>
  <footer id="contact">
    <p>
      &copy; 2026 My Styled Webpage | Designed using HTML5 and CSS3
    </p>
  </footer>
</body>
</html>
```

```

</p>
</footer>
<script>
const canvas = document.getElementById("myCanvas");
const context = canvas.getContext("2d");

context.fillStyle = "skyblue";
context.fillRect(40, 50, 180, 100);

context.beginPath();
context.arc(350, 100, 50, 0, 2 * Math.PI);
context.fillStyle = "orange";
context.fill();

context.beginPath();
context.moveTo(450, 200);
context.lineTo(650, 80);
context.strokeStyle = "green";
context.lineWidth = 5;
context.stroke();

context.font = "bold 30px Arial";
context.fillStyle = "purple";
context.fillText("Kongarani Sirisha", 220, 300);
</script>
<style>
* {
margin: 0;
padding: 0;
box-sizing: border-box;
}
body {
font-family: Arial, Helvetica, sans-serif;
background-color: #e1e5ec;
color: #333333;
line-height: 1.9;
}
header {
background: linear-gradient(90deg, #1e3c72, #71c1dc);
color: rgb(237, 237, 237);
text-align: center;
padding: 35px;
text-shadow: 2px 2px 5px black;
}
header h1 {
font-family: Georgia, serif;
font-size: 42px;
}
header p {
font-size: 20px;
margin-top: 10px;
}
nav {
background-color: #111;
text-align: center;
padding: 15px;
}
nav a {
color: white;
text-decoration: none;
font-size: 18px;
margin: 0 20px;
padding: 8px 12px;
border-radius: 6px;

```

```
transition: 0.3s;
}
nav a:hover {
background-color: rgb(255, 255, 255);
color: #1e3c72;
}
main {
width: 90%;
max-width: 1100px;
margin: 30px auto;
}
section {
margin-bottom: 50px;
}
section h2 {
text-align: center;
font-family: Georgia, serif;
font-size: 30px;
color: #1e3c72;
margin-bottom: 25px;
}
canvas {
display: block;
max-width: 100%;
margin: auto;
background-color: white;
border: 4px solid #1e3c72;
border-radius: 15px;
padding: 10px;
box-shadow: 0 6px 18px rgba(0, 0, 0, 0.25);
}
.card-container {
display: flex;
justify-content: center;
gap: 25px;
flex-wrap: wrap;
}
.card {
width: 300px;
min-height: 260px;
background: linear-gradient(135deg, white, #dbeafe);
border: 2px solid #2a5298;
border-radius: 20px;
margin: 10px;
padding: 25px;
text-align: center;
box-shadow: 0 8px 18px rgba(0, 0, 0, 0.2);
transition: 0.3s;
}
.card:hover {
transform: translateY(-10px);
box-shadow: 0 14px 25px rgba(0, 0, 0, 0.3);
}
.card h3 {
color: #1e3c72;
font-size: 25px;
margin-bottom: 15px;
}
.card p {
font-family: Verdana, sans-serif;
font-size: 16px;
margin-bottom: 20px;
}
button {
background-color: #1e3c72;
```

```
color: white;
font-size: 16px;
padding: 10px 18px;
border: none;
border-radius: 25px;
cursor: pointer;
transition: 0.3s;
}
button:hover {
background-color: #f97316;
transform: scale(1.1);
box-shadow: 0 5px 12px rgba(0, 0, 0, 0.3);
}
footer {
background-color: #111;
color: white;
text-align: center;
padding: 25px;
font-size: 17px;
}
</style>
</body>
</html>
```

7. CODE EXPLANATION

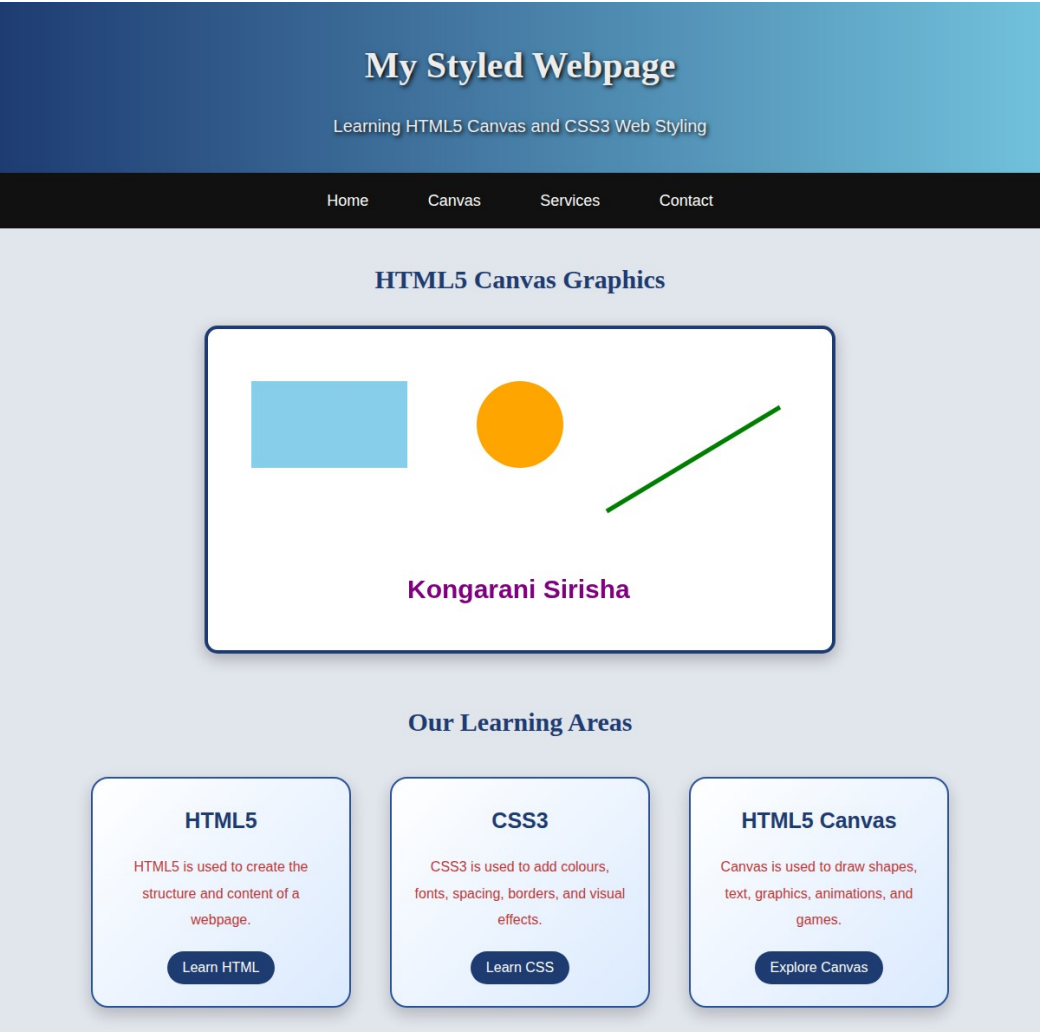
Tag / Attribute	Explanation
<canvas id="myCanvas">	Reserves a drawable area on the page.
getContext("2d")	Returns the 2D drawing context used to draw shapes.
fillRect(x,y,w,h) / strokeRect(x,y,w,h)	Draws a filled or outlined rectangle at the given position and size.
arc(x,y,r,0,2*PI) + fill()/stroke()	Draws a circle of radius r centred at (x, y).
moveTo() / lineTo()	Draws a straight line between two points.
fillText()	Renders text (the student's name) onto the canvas.
linear-gradient()	Creates a smooth gradient background for the header and cards.
box-shadow	Adds a soft drop shadow around the canvas and content cards.
text-shadow	Adds a shadow behind the header title text.
border-radius	Rounds the corners of the canvas, cards and buttons.
:hover + transition	Animates nav links, cards and buttons smoothly when the mouse hovers over them.

8. TEST CASES AND EXECUTION DATA

The webpage was opened in a browser and each section was verified against its expected behaviour. The output for each test case is shown right below it.

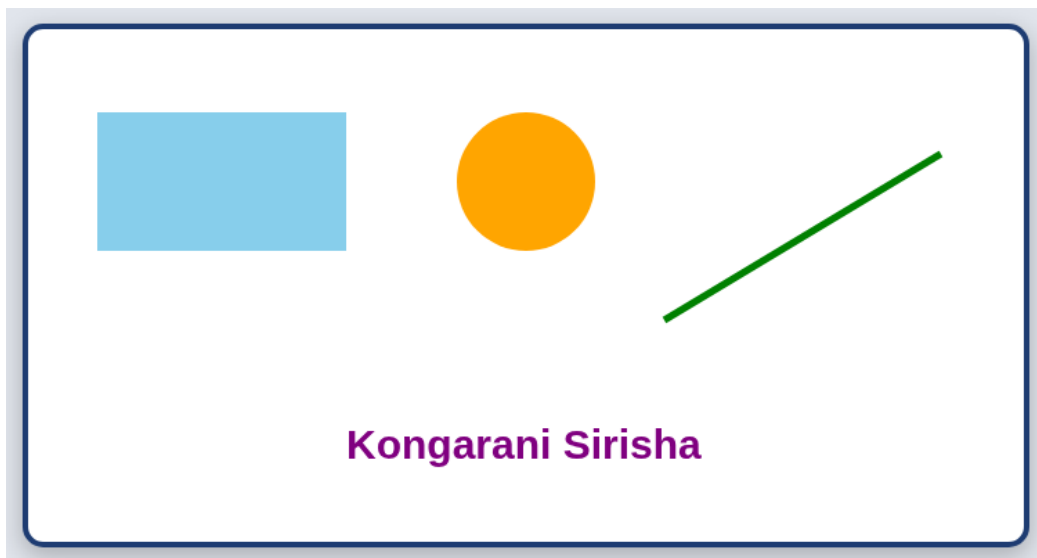
Test Case	TC-01
Action / Input	Load the page in browser
Expected Result	Header, navigation bar, canvas and cards display correctly
Actual Result	Displayed correctly
Status	Pass

TC-01 Output



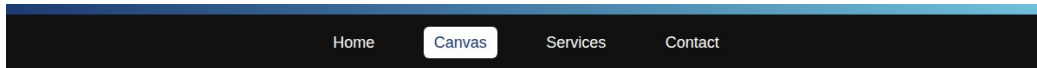
Test Case	TC-02
Action / Input	Inspect the canvas area
Expected Result	Rectangle, circle, line and name text are visible
Actual Result	All four elements rendered
Status	Pass

TC-02 Output



Test Case	TC-03
Action / Input	Hover over a navigation link
Expected Result	Link background changes with a smooth transition
Actual Result	Background changed smoothly
Status	Pass

TC-03 Output



Test Case	TC-04
Action / Input	Hover over a content card
Expected Result	Card lifts up with an enhanced shadow
Actual Result	Lift and shadow effect worked
Status	Pass

TC-04 Output



Test Case	TC-05
Action /	Hover over a button

Input	
Expected Result	Button color changes and scales slightly
Actual Result	Color changed and button scaled
Status	Pass

TC-05 Output



Test Case	TC-06
Action / Input	Resize the browser window
Expected Result	Cards wrap responsively without overlapping
Actual Result	Cards wrapped correctly
Status	Pass

TC-06 Output

HTML5

HTML5 is used to create the structure and content of a webpage.

[Learn HTML](#)

CSS3

CSS3 is used to add colours, fonts, spacing, borders, and visual effects.

[Learn CSS](#)

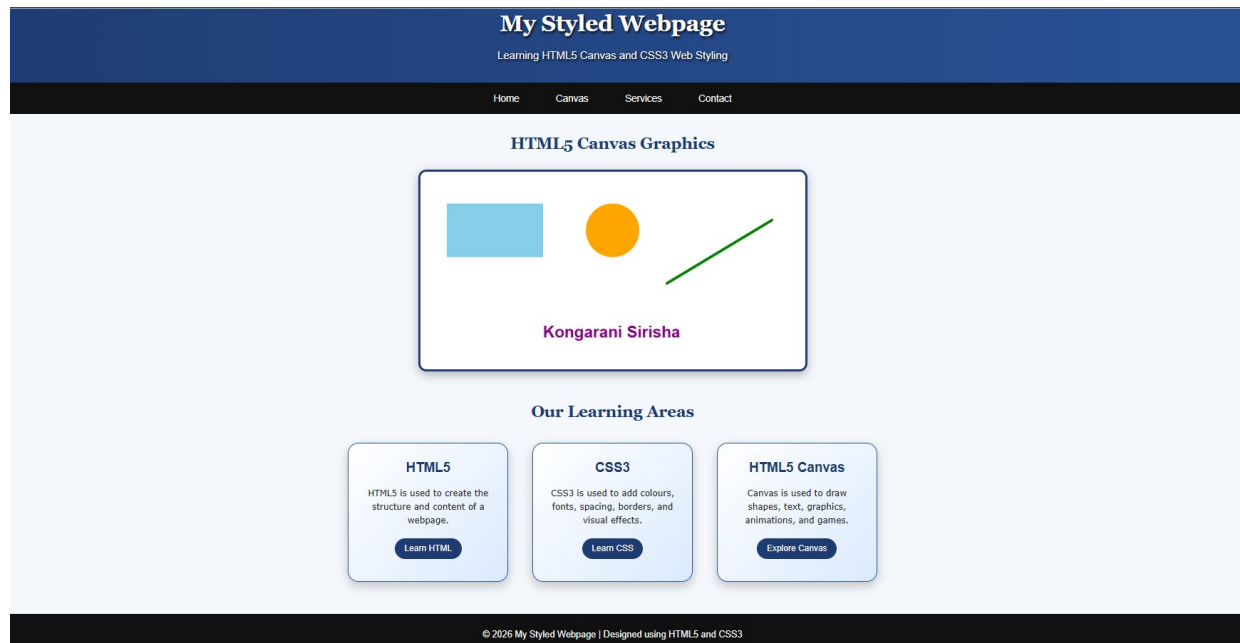
HTML5 Canvas

Canvas is used to draw shapes, text, graphics, animations, and games.

[Explore Canvas](#)

9. OUTPUT

Output — Header, Navigation Bar, HTML5 Canvas Graphics, Content Cards and Footer



10. RESULT

Thus, a webpage titled "My Styled Webpage" was successfully developed, demonstrating HTML5 Canvas drawing (rectangle, circle, line and text) along with CSS3 styling features such as gradients, shadows, rounded corners and hover transition effects. The output was verified against the expected results for all test cases.